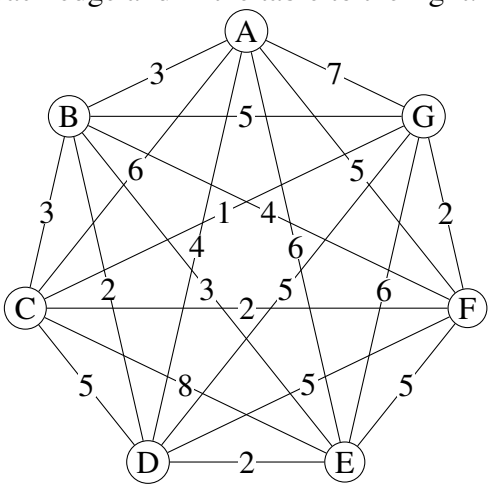
Name: Solutions score:_____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

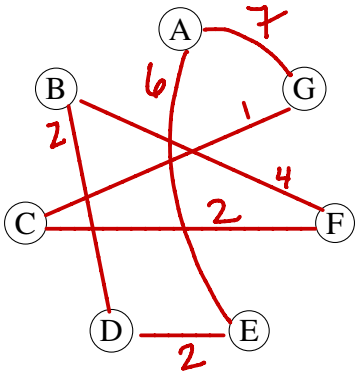
Questions 1 and 2 will use the weighted graph below. Edges weights are written in the middle of each edge and in the table to the right.



	A	B	C	D	E	F	G
A	0	3	6	4	6	5	~~7~~
B	3	0	3	2	3	4	5
C	6	3	0	5	8	2	~~X~~
D	4	~~2~~	5	0	2	5	5
E	6	3	8	2	0	5	6
F	5	4	~~8~~	5	5	0	~~8~~
G	7	5	1	5	6	2	0

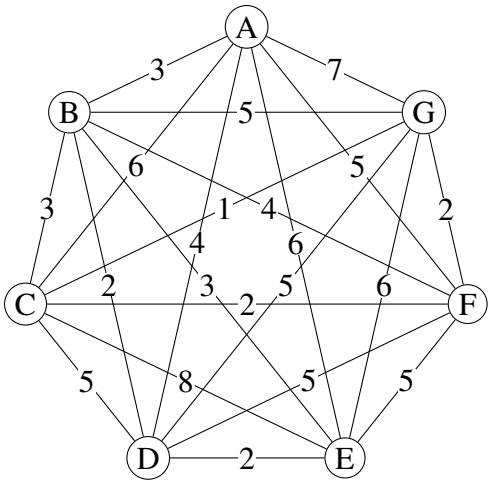
1. (4 points) Use the Nearest Neighbor Algorithm (NNA) starting at vertex G to find a Hamiltonian circuit in the graph. Draw the circuit on the vertices below and determine weight of the circuit. The weights of the edges are listed in increasing order in the table above.

   Nearest Neighbor Algorithm: From the present vertex, move to the nearest **unvisited** vertex until all vertices are visited. Then close the circuit.



1+2+2+2+4+6+7 = 24

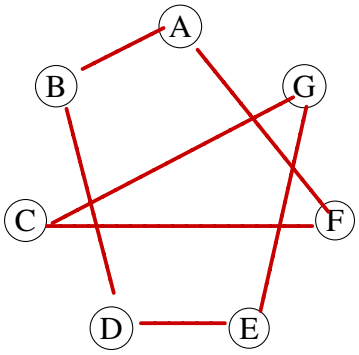
circuit weight: 24.



edge	weight	edge	weight	edge	weight
CG	1 ✓	~~BE~~	~~3~~	DG	5
BD	2 ✓	~~AD~~	~~4~~	EF	5
CF	2 ✓	~~BF~~	~~4~~	AC	6
DE	2 ✓	AF	5 ✓	AE	6
~~FG~~	~~2~~	BG	5	EG	6
AB	3 ✓	CD	5	AG	7
~~BC~~	~~3~~	DF	5	CE	8

2. (4 points) Use the Cheapest Link Algorithm to find a Hamiltonian circuit in the graph (drawn above). Draw the circuit on the vertices below and determine weight of the circuit.

Cheapest Link Algorithm: Select the cheapest edge in the graph that does not create a vertex of degree 3 or close the circuit too soon.



1+2+2+2+3+5+6

= 10 + 10 + 1

circuit weight: 21.

3. (1 point) How many different Hamiltonian circuits does a graph on 7 vertices have? (It is sufficient to indicate how it is computed.)

$\frac{6!}{2}$ or 6! w/o symmetry

4. (1 point) Is either algorithm, NNA or the Cheapest Link, **guaranteed** to give the cheapest Hamiltonian circuit in **every** graph? Explain your answer.

No. In this example NNA from G did not get a cheapest circuit. There are examples where Cheapest Link does very badly.

